

Stony Brook University
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ASSIGNMENT 5

SOLUTIONS

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you used. The last problem is optional, if you attempt it you can get extra credit.

1. You have a data set consisting on a cross-sample of 4131 high-school students. The variables you have are: *sat*, which is the combined SAT score; *hsize* is the size of the student's high school graduating class, in hundreds; *female* is a gender dummy variable, having a value of one if female or zero if male; and *black* which is a race dummy variable equal to one for blacks and zero otherwise.

	<i>Dependent variable:</i>	
	<i>sat</i>	
	(1)	(2)
<i>hsize</i>	19.297*** (3.817)	3.933*** (1.178)
<i>hsize</i> ²	-2.195*** (0.524)	
<i>hsize</i> ² - 7 <i>hsize</i>		-2.195*** (0.524)
<i>female</i>	-45.091*** (4.230)	-45.091*** (4.230)
<i>black</i>	-169.813*** (15.434)	-169.813*** (15.434)
<i>female</i> × <i>black</i>	62.306*** (19.320)	62.306*** (19.320)
Constant	1,028.097*** (6.360)	1,028.097*** (6.360)
Observations	4,131	4,131
<i>R</i> ²	0.086	0.086
Adjusted <i>R</i> ²	0.085	0.085
Residual Std. Error	133.369	133.369
Overall F Statistic (df = 5; 4131)	77.524***	77.524***
<i>F</i> Statistic <i>black</i> + <i>female</i> × <i>black</i> = 0 (df=1;4131)	84.965***	
Note: *p<0.1; **p<0.05; ***p<0.01		

- (a) Consider the model in column (1). Write down the population model that the researcher has in mind.
- (b) Is there strong evidence that $hsize^2$ should be included in the model?
- (c) Consider student A , which graduated from a high school class of 300 ($hsize = 3$), is female and not black. Student B graduated from a high school class of 400 ($hsize = 4$) and is also female and not black. Let $\Delta sat = sat_B - sat_A$ denote the difference in SAT scores where sat_B and sat_A are the SAT scores of students B and A respectively. Rewrite the regression model so that you can construct a 95% confidence interval around Δsat . Compute this confidence interval. *Hint: You will have to use the regression in model (2). How do you rewrite the regression model so that you can use it?*
- (d) Holding the value of $hsize$ fixed, what is the estimated difference in SAT score between black females and nonblack females? State the null hypothesis that you would need to test to determine if the difference is statistically significant. Test it at the 5% level. After you test this null hypothesis, construct a 95% confidence interval around the estimated difference. *Hint: Look at the F statistic at the bottom of the table. What is its meaning? To construct the confidence interval, you will have to use one of the formulas we saw in class.*

Solution

- (a) The model that is estimated in column (1) is

$$sat_i = \beta_0 + \beta_1 hsize_i + \beta_2 hsize_i^2 + \beta_3 female_i + \beta_4 black_i + \beta_5 female_i \times black_i + u_i.$$

- (b) The hypothesis test to answer this question is $H_0 : \beta_2 = 0$ vs $H_a : \beta_2 \neq 0$. The t statistic is -4.18 which in absolute value is well above the critical values at the 10%, 5% or 1% levels (1.64, 1.96 and 2.58 respectively). We could also see this from the p -value whose results are indicated as stars in the table: three stars (***) means that the p -value is smaller than 0.01, thus we reject the null at any level equal or above 1%. Therefore, $hsize^2$ should be included in the model.
- (c) We have that

$$\begin{aligned} sat_A &= E[sat | hsize = 300, female = 1, black = 0] = \beta_0 + 3\beta_1 + 3^2\beta_2 + \beta_3, \\ sat_B &= E[sat | hsize = 400, female = 1, black = 0] = \beta_0 + 4\beta_1 + 4^2\beta_2 + \beta_3, \\ \Delta sat &= sat_B - sat_A = \beta_1 + 7\beta_2. \end{aligned}$$

From the estimates in model (1), the previous change in SAT score can be estimated as $\widehat{\Delta sat} = \hat{\beta}_1 + 7\hat{\beta}_2 = 3.933$. This number appears as the estimated coefficient of $hsize$ in model (2). This points out at a connection between them.

Let's see how we can rewrite the model so that we can use model (2) to construct the confidence interval around the change. To the equation in model (1), we will add and subtract $7\beta_2 hsize$:

$$\begin{aligned} sat_i &= \beta_0 + \beta_1 hsize_i + 7\beta_2 hsize_i - 7\beta_2 hsize_i + \beta_2 hsize_i^2 + \beta_3 female_i + \beta_4 black_i + \beta_5 female_i \times black_i + u_i, \\ &= \beta_0 + (\beta_1 + 7\beta_2) hsize_i + \beta_2 (hsize_i^2 - 7hsize_i) + \beta_3 female_i + \beta_4 black_i + \beta_5 female_i \times black_i + u_i, \\ &= \beta_0 + \theta_1 hsize_i + \beta_2 R_i + \beta_3 female_i + \beta_4 black_i + \beta_5 female_i \times black_i + u_i, \end{aligned}$$

where $\theta_1 = \beta_1 + 7\beta_2$ and $R_i = hsize_i^2 - 7hsize_i$. This is precisely the equation that model (2) is estimating. The estimated coefficient $\hat{\theta}_1$ is precisely the change we are looking for and its standard error of 1.178 is the one we need. Thus, for a 95% confidence interval, we have that $[3.933 \pm 1.96 \times 1.178] = [1.624, 6.242]$.

- (d) The average difference in SAT scores between a female black student vs a female non-black student who attended high schools with the same size is given by (after simplifying all terms):

$$E[\text{sat}|\text{hsize}, \text{female} = 1, \text{black} = 1] - E[\text{sat}|\text{hsize}, \text{female} = 1, \text{black} = 0] = \beta_4 + \beta_5.$$

This difference is estimated as $\hat{\beta}_4 + \hat{\beta}_5 = -107.507$.

We need to test for this difference being statistically different from zero. To do this, we state the hypothesis $H_0 : \beta_4 + \beta_5 = 0$ vs $H_a : \beta_4 + \beta_5 \neq 0$. To test this hypotheses, we need the appropriate F statistic. Notice that the bottom of the table provides precisely this number. In there we can see that the $F = 84.965$. Thus, the standard error we are looking for is given by

$$SE(\hat{\beta}_4 + \hat{\beta}_5) = \frac{|-107.507|}{\sqrt{84.965}} = 11.6632.$$

Therefore, the 95% confidence interval is $[-107.507 \pm 1.96(11.6632)] = [-130.3669, -84.6471]$.

2. This problem is inspired by a study of the “gender gap” in earnings in top corporate jobs (Bertrand, M. and Hallock, K.F., 2001. *The gender gap in top corporate jobs. Industrial & Labor Relations Review*, 55(1), pp.3–21). The study compares total compensation among top executives in a large set of U.S. public corporations in the 1990s.

- (a) Let *Female* be an indicator variable that is equal to 1 for females and 0 for males. A regression of the logarithm of earnings onto *Female* yields

$$\widehat{\ln(\text{Earnings})} = \underset{(0.01)}{6.48} - \underset{(0.05)}{0.44} \text{Female}, \text{SER} = 2.65.$$

- The estimated coefficient on *Female* is -0.44 . Explain what this value means.
 - The *SER* is 2.65. Explain what this value means.
 - Does this regression suggest that female top executives earn less than top male executives? Explain.
 - Does this regression suggest that there is gender discrimination?
- (b) Two new variables, the market value of the firm (a measure of firm size, in millions of dollars) and stock return (a measure of firm performance, in percentage points), are added to the regression:

$$\widehat{\ln(\text{Earnings})} = \underset{(0.03)}{3.86} - \underset{(0.04)}{0.28} \text{Female} + \underset{(0.004)}{0.37} \ln(\text{MarketValue}) + \underset{(0.003)}{0.004} \text{Return}, n = 46670, \bar{R}^2 = 0.345$$

- The coefficient on $\ln(\text{MarketValue})$ is 0.37. Explain what this value means.
- (c) Are large firms more likely than small firms to have female top executives? Explain.

Solution

- (a)
- $\ln(\text{Earnings})$ for females are, on average, 0.44 lower for men than for women.
 - The error term has a standard deviation of 2.65 (measured in log-points).
 - Yes. But the regression does not control for many factors (size of firm, industry, profitability, experience and so forth).
 - No. In isolation, these results do not imply gender discrimination. Gender discrimination means that two workers, identical in every way but gender, are paid different wages. Thus, it is also important to control for characteristics of the workers that may affect their productivity (education, years of experience, etc.) If these characteristics are systematically different between men and women, then they may be responsible for the difference in mean wages. (If this were true, it would raise an interesting and important question of why women tend to have less education or less experience than men, but that is a question about something other than gender discrimination.)

nation.) These are potentially important omitted variables in the regression that will lead to bias in the OLS coefficient estimator for Female. Since these characteristics were not controlled for in the statistical analysis, it is premature to reach a conclusion about gender discrimination.

- (b) If *MarketValue* increases by 1%, earnings increase by 0.37%
- (c) Forgetting about the effect of *Return*, whose effects seems small and statistically insignificant, the omitted variable bias formula suggests that Female is negatively correlated with $\ln(\text{MarketValue})$.

3. Consider the regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{1i} \times X_{2i} + u_i.$$

Calculate

- (a) $\frac{\Delta Y_i}{\Delta X_1}$ (effect of a change in X_1 holding X_2 constant).
- (b) $\frac{\Delta Y_i}{\Delta X_2}$ (effect of a change in X_2 holding X_1 constant).
- (c) If X_1 changes by ΔX_1 and X_2 changes by ΔX_2 , calculate ΔY .

Solution

- (a) $\frac{\Delta Y_i}{\Delta X_1} = \beta_1 + \beta_3 X_{2i}$.
- (b) $\frac{\Delta Y_i}{\Delta X_2} = \beta_2 + \beta_3 X_{1i}$.
- (c) We have

$$\begin{aligned} Y_i + \Delta Y &= \beta_0 + \beta_1(X_{1i} + \Delta X_1) + \beta_2(X_{2i} + \Delta X_2) + \beta_3((X_{1i} + \Delta X_1) \times (X_{2i} + \Delta X_2)) \\ &= \beta_0 + \beta_1(X_{1i} + \Delta X_1) + \beta_2(X_{2i} + \Delta X_2) \\ &\quad + \beta_3(X_{1i} \times X_{2i} + X_{1i} \times \Delta X_2 + \Delta X_1 \times X_{2i} + \Delta X_1 \times \Delta X_2), \end{aligned}$$

Then, after we subtract $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{1i} \times X_{2i}$ we get

$$\Delta Y = \beta_1 \Delta X_1 + \beta_2 \Delta X_2 + \beta_3(X_{1i} \times \Delta X_2 + \Delta X_1 \times X_{2i} + \Delta X_1 \times \Delta X_2)$$

4. Use the data in BWGHT2 for this exercise.

- (a) Estimate the equation

$$\log(\text{bwght}) = \beta_0 + \beta_1 \text{npvis} + \beta_2 \text{npvis}^2 + u$$

by OLS, and report the results in the usual way. Is the quadratic term significant?

- (b) Show that, based on the equation from part (a), the number of prenatal visits that maximizes $\log(\text{bwght})$ is estimated to be about 22. How many women have at least 22 prenatal visits in the sample?
- (c) Does it make sense that birth weight is actually predicted to decline after 22 prenatal visits? Explain.
- (d) Add mother's age to the equation, using a quadratic functional form. Holding *npvis* fixed, at what mother's age is the birth weight of the child maximized? What fraction of women in the sample are older than the "optimal" age? *Hint: the quadratic equation $y = A + Bx + Cx^2$ defines a parabola, where A, B and C are constants. If $C < 0$ then the parabola has a global maximum at $x^* = -\frac{B}{2C}$.*

- (e) Would you say that mother's age and number of prenatal visits explain a lot of the variation in $\log(bwght)$?
- (f) Using quadratics for both $npvis$ and age , decide whether using the natural log or the level of $bwght$ is better for predicting $bwght$.
5. (Extra Credit) Consider a model where the return to education depends upon the amount of work experience (and viceversa):

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 educ \times exper + u$$

- (a) Show that the return to another year of education (in decimal form), holding $exper$ fixed is $\beta_1 + \beta_3 exper$.
- (b) State the null hypothesis that the return to education does not depend on the level of $exper$. What do you think is the appropriate alternative?
- (c) use the data in WAGE2 to test the null hypothesis in (b) against your stated alternative.
- (d) Let θ_1 denote the return to education (in decimal form), when $exper = 10$: $\theta_1 = \beta_1 + 10\beta_3$. Obtain $\hat{\theta}_1$ and a 95% confidence interval for θ_1 . *Hint: Rewrite the regression. Here is a useful trick to do it: write $\beta_1 = \theta_1 - 10\beta_3$ and plug this into the equation; then rearrange.*